Week12:Develop android application to send messages to mobile device by using Fire based cloud messaging and push notifications

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fire based cloud messaging Applocation and Push Notifications"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button
        android:id="@+id/btnSendNotification"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"/>

</LinearLayout>
```

mainactivity.java

```
package com.example.notificationapp;

import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;

public class MainActivity extends AppCompatActivity {

    private static final String CHANNEL_ID = "notification_channel";
    private static final int NOTIFICATION_PERMISSION_CODE = 100;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        // Create notification channel
        createNotificationChannel();
```

```
    // Ask notification permission for Android 13+
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        if (ActivityCompat.checkSelfPermission(
                this,
                Manifest.permission.POST_NOTIFICATIONS
        ) != PackageManager.PERMISSION_GRANTED) {

            ActivityCompat.requestPermissions(
                    this,
                    new String[]{
                            Manifest.permission.POST_NOTIFICATIONS
                    },
                    NOTIFICATION_PERMISSION_CODE
            );
        }
    }

    // Send Notification button
    findViewById(R.id.btnSendNotification)
            .setOnClickListener(v -> sendNotification());
}

private void createNotificationChannel() {

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

        NotificationChannel channel =
                new NotificationChannel(
                        CHANNEL_ID,
                        "My Notifications",
                        NotificationManager.IMPORTANCE_HIGH
                );

        channel.setDescription("Application notifications");

        NotificationManager manager =
                getSystemService(NotificationManager.class);

        manager.createNotificationChannel(channel);
    }
}

private void sendNotification() {

    // Android 13+ permission check
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {

        if (ActivityCompat.checkSelfPermission(
                this,
                Manifest.permission.POST_NOTIFICATIONS
        ) != PackageManager.PERMISSION_GRANTED) {

            return;
        }
    }

    NotificationCompat.Builder builder =
            new NotificationCompat.Builder(this, CHANNEL_ID)
                    .setSmallIcon(R.mipmap.ic_launcher)
                    .setContentTitle("New Message")
```

```
                .setContentText("Hello! This is a test notification.")
                .setPriority(NotificationCompat.PRIORITY_HIGH)
                .setAutoCancel(true);

        NotificationManagerCompat notificationManager =
                NotificationManagerCompat.from(this);

        notificationManager.notify(1, builder.build());
    }
}
```

Androidmanifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission
        android:name="android.permission.POST_NOTIFICATIONS" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.NotificationApp">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:windowSoftInputMode="adjustResize">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```